

Micro I, class 21

The main point was the Debreu-Fan-Glicksberg theorem on the existence of Nash equilibrium. The details are in my notes on the subject.

I then wrote out, as a Corollary to the D-F-G theorem, Nash's 1950 and 1951 theorems on existence of a Nash equilibrium for a mixed extension of a finite game. (Of course he didn't call it a *Nash equilibrium*, though he was far from a modest man; he just referred to it as an *equilibrium point*.)

I illustrated it through the use of “nice games,” with compact interval strategy sets, and continuous utility representations that are strictly quasiconcave in own strategies. For two player games, a Nash equilibrium exists if the function $f(s_1) = r_1(r_2(s_1))$ from S_1 into S_1 has a fixed point, that is, a point $s_1^* \in S_1$ with $f(s_1^*) = s_1^*$. A Nash equilibrium is the pair $(s_1^*, r_2(s_1^*))$. Since, under these assumptions, each best reply is a continuous function, the conclusion follows immediately from the elementary *Intermediate Value Theorem*, a very special case of Kakutani's fixed point theorem, which is used in the proof of the D-F-G theorem, as I hinted at the “small type” discussion in my notes.

Next class I will turn to Nash equilibria of Cournot games. I have posted my notes on the Cournot game seen through the lens of the D-F-G theorem. And after I will turn to the *Bertrand game* of firms competing in prices.